3. Multi-Factor Authentication (MFA)

Multi-Factor Authentication (MFA) requires users to verify their identity using two or more authentication factors.

Authentication Factors

1. **Something you know**
    - Password
    - PIN
2. **Something you have**
    - Mobile phone
    - Hardware security key
    - Authenticator app
3. **Something you are**
    - Fingerprint
    - Face recognition
    - Iris scan

Example

A user signs in with:

- Password ✓
- One-Time Password (OTP) from an authenticator app ✓

Access is granted only after both checks succeed.

Benefits

- Reduces risk of account compromise
- Protects against stolen passwords
- Improves security for remote access
- Helps meet compliance requirements

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4. Password Policy

A password policy defines rules for creating and managing passwords.

Common Requirements

- Minimum length (e.g., 12 characters)
- Uppercase letters
- Lowercase letters
- Numbers
- Special characters
- No common or easily guessed passwords

Example of a Strong Password

MyComp@ny2026!

Password Management Best Practices

- Use unique passwords for different accounts.
- Avoid sharing passwords.
- Do not write passwords where others can access them.
- Use a trusted password manager when appropriate.
- Change passwords immediately if compromise is suspected.
- Enable MFA wherever possible.

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5. Security Best Practices

- Lock your computer when away from your desk (Windows + L).
- Keep Windows and applications updated.
- Be cautious with email attachments and links.
- Install software only from trusted sources.
- Use least-privilege access (only the permissions required).

- Back up important data regularly.
- Report suspicious activity or phishing attempts immediately.

Common Security Threats

Threat	Description	Prevention
Virus	Malicious program that infects files	Antivirus, regular scans
Ransomware	Encrypts files and demands payment	Backups, antivirus, updates
Phishing	Fake emails or websites to steal credentials	Verify senders, don't click suspicious links
Spyware	Secretly collects user information	Antivirus, trusted software
Brute-force attack	Repeated password guessing	Strong passwords, MFA, account lockout

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IT Support Responsibilities

As an IT Support Engineer, you may:

- Install and update antivirus software.
- Verify BitLocker encryption status.
- Help users recover devices with BitLocker recovery keys (following company procedures).
- Assist users with MFA setup or troubleshooting.
- Reset passwords according to company policy.
- Remove malware and ensure systems are fully updated.
- Educate users about phishing and safe security practices.

Interview Questions

Q1. What is the purpose of antivirus software?
Answer: It detects, prevents, and removes malware such as viruses, ransomware, spyware, and trojans to protect systems and data.

Q2. What is BitLocker?
Answer: BitLocker is Microsoft's full-disk encryption technology that protects data on Windows devices by encrypting the drive.

Q3. What is a BitLocker recovery key?
Answer: It is a 48-digit key used to unlock an encrypted drive when normal startup verification fails.

Q4. What is Multi-Factor Authentication (MFA)?
Answer: MFA requires two or more forms of authentication, such as a password plus a one-time code, to verify a user's identity.

Q5. Why is MFA more secure than using only a password?
Answer: Even if a password is stolen, an attacker still needs the second authentication factor, making unauthorized access much harder.

Q6. What makes a strong password?
Answer: A strong password is long, unique, difficult to guess, and combines uppercase and lowercase letters, numbers, and special characters.

Q7. What should you do if you receive a suspicious email?
Answer: Do not click links or open attachments. Report the email to the IT/security team and verify its legitimacy before taking any action.